SEM Image Restoration

Joint Denoising & 2× Super-Resolution – Technical Report

KLA AI Hackathon 2026

Team: ECE_Lords

A complete account of the data analysis, architecture and loss-function selection, production training, and post-training experiments behind the submitted model.

At a glance

Dataset	4,785 GT/NoisyLR pairs (4,278 train / 477 val)
Architecture	NAFNet-full, 29.07M parameters
Training	600 epochs, 5.22 hours, RTX 4050
Final result	PSNR 23.64 dB / SSIM 0.625 / LPIPS 0.136
Improvement vs. bicubic baseline	+3.42 dB PSNR, +0.122 SSIM

1 Problem Statement

The task: restore SEM (scanning-electron-microscopy) images degraded by speckle noise, Gaussian noise, and a 2× spatial-resolution reduction, recovering both signal quality and full resolution. Evaluation uses three metrics – **PSNR**, **SSIM**, **LPIPS** – plus inference latency. Models must also generalize to out-of-distribution test samples.

- **Two project phases:** an earlier phase used a smaller (3,200-pair) dataset; the current phase received a larger, harder, differently-distributed dataset (`semicon_train_data`, 4,785 pairs). This report documents the current phase in full, using the earlier model only as a comparison baseline (Section 6).
- All design decisions below trace back to a specific measured finding from the data itself – not a default choice or a literature assumption carried over unchecked.

2 Exploratory Data Analysis

Full-population analysis – every one of the 4,785 pairs was processed, not a sample.

> **Why this matters:** every architectural and loss-function decision in Sections 4–6 is a direct response to one of the findings below. This is the evidentiary foundation of the whole model design, not a preliminary formality.

Integrity: 4,785 clean pairs, zero corrupt files, zero duplicates – no special-casing needed in the data loader.

Intensity range: GT is strictly $[0, 1]$; NoisyLR overshoots to roughly $[-0.31, 2.24]$. 99.1% of images have at least one pixel > 1; the overshoot is concentrated in *bright* regions (mean GT intensity 0.778 where it occurs) and undershoot in *dark* regions (mean GT intensity 0.063) – i.e., it is content-dependent, not a fixed offset. **Consequence:** never clip the network's input; clip only the final output to $[0, 1]$.

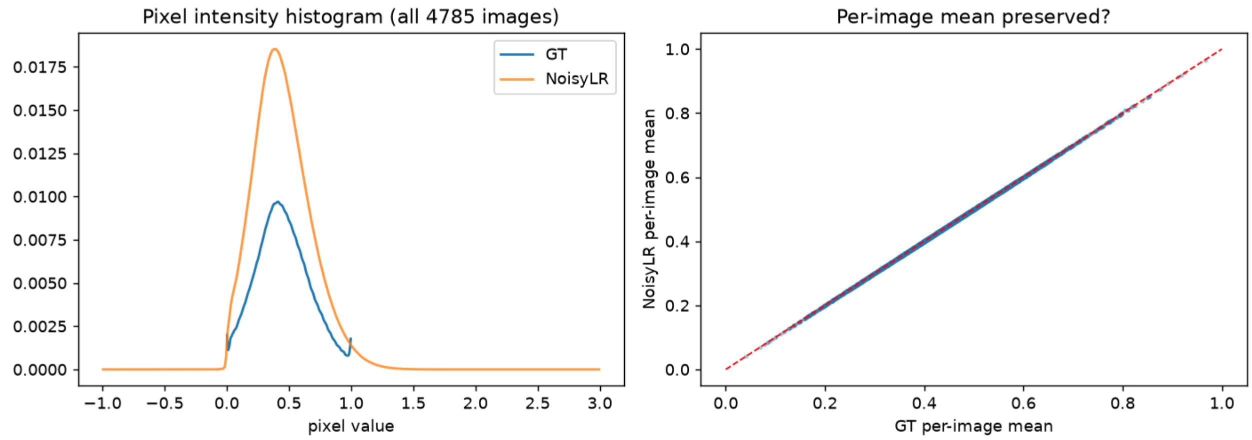

Figure 1: Intensity overshoot/undershoot behaviour – content-dependent, not a fixed offset.

Left: the pixel-value histogram of GT (blue) vs. NoisyLR (orange) across all 4,785 images – NoisyLR's much wider spread is the speckle noise pushing values outside $[0, 1]$. *Right: per-image mean intensity, GT vs. NoisyLR – points sit almost exactly on the diagonal, confirming the noise does not shift overall brightness, only spread.*

Noise is signal-dependent (speckle-like), not uniform: residual variance rises from 0.00041 in the darkest intensity bin to 0.0323 in the brightest, a clear positive slope (0.0356). A loss that assumes constant noise variance is measurably mismatched with this data. The downsampling kernel is also confirmed **bicubic-like** here (MSE 0.0106 vs. an area-average reference, 0.0105 vs. a bicubic reference) – this licenses the residual-over-bicubic-baseline architecture used in Section 4.

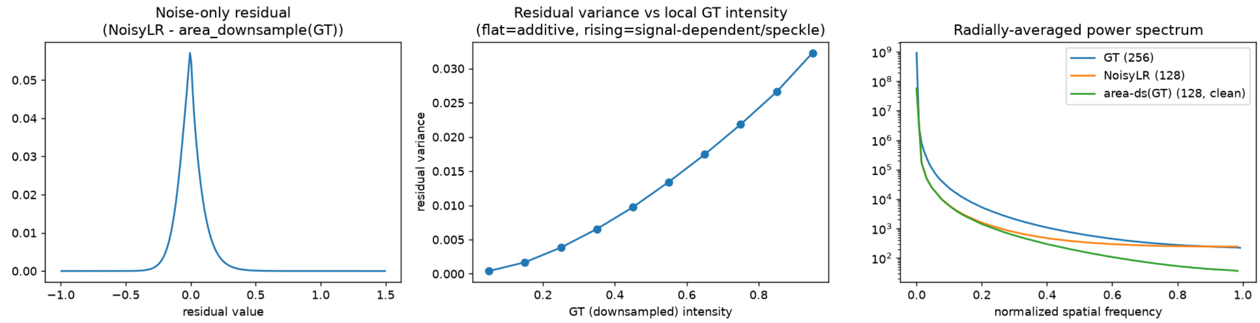

Figure 2: Signal-dependent (speckle-like) noise: residual variance rises with local intensity.

Left: the raw noise-only residual (NoisyLR minus a clean downsample of GT) is sharply peaked at zero – unbiased noise. Middle: binning that residual by local GT brightness shows variance climbing steadily from dark to bright regions – a flat line would mean simple additive noise; this rising line is the direct evidence for signal-dependent, speckle-like noise. Right: radially-averaged frequency spectrum, showing the noisy signal's high-frequency floor sitting above the clean reference.

Real detail is lost, not just masked by noise: Sobel gradient magnitude is *higher* in the degraded image (1.43×, noise artifact), but Laplacian variance – a stronger true-structure detector – is *lower* by 2.9× (ratio 0.350). The two disagreeing is itself the finding: the network must both denoise *and* synthesize real detail, which a plain pixel loss cannot target on its own. This directly motivated the dedicated edge/Laplacian loss term (Section 4.3).

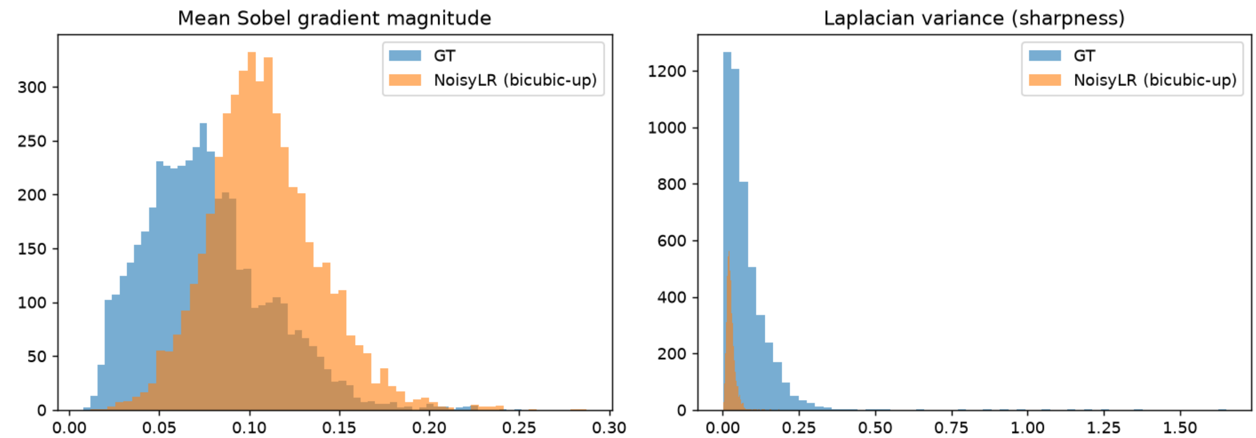

Figure 3: Real detail loss vs. noise-inflated gradients: Sobel vs. Laplacian disagree.

Left: Sobel gradient magnitude is actually higher *in the degraded image (orange, shifted right of GT's blue) – but this is a false signal, since noise itself creates sharp pixel-to-pixel jumps. Right: Laplacian variance – a stronger true- structure detector – tells the opposite story: GT (blue) has a long tail of genuinely sharp content that the degraded image (orange, collapsed near zero) has lost. The two metrics disagreeing is the actual finding.*

Corrupted ground truth exists and must be excluded. A block-wise spatial-autocorrelation detector (max lag-1 correlation over 32×32 blocks – refined after an earlier whole-image version produced false positives on real, sparse-content images) found **30 confirmed-corrupted images (0.63%)**, visually verified, not just score-thresholded. Excluded from both splits.

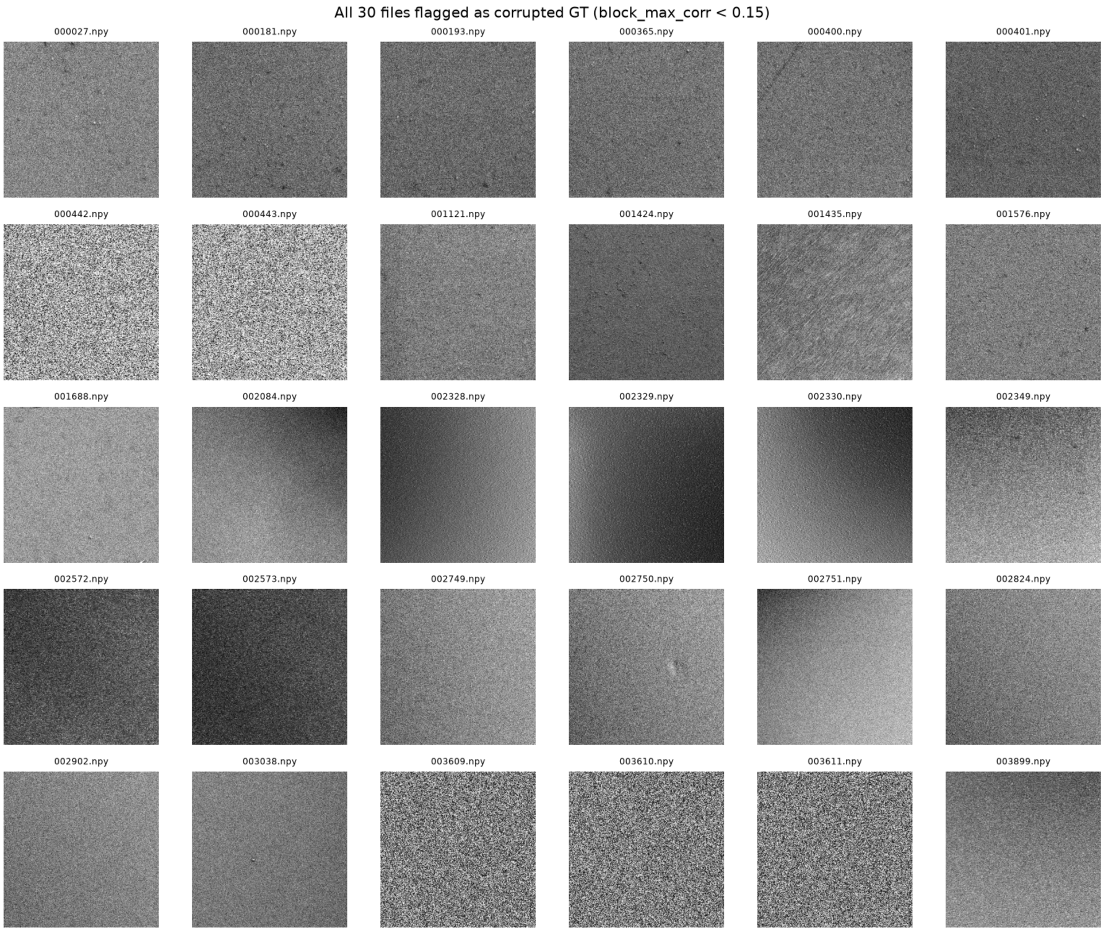

Figure 4: The 30 confirmed corrupted-ground-truth images, excluded from training and validation.

Every one of the 30 images flagged by the block-autocorrelation detector and visually confirmed as corrupted – pure sensor static with no coherent surface geometry anywhere in the frame, unlike genuinely grainy-but-real high-noise acquisitions elsewhere in the dataset. These were excluded from both the training and validation splits.

Content forms ∼10 texture clusters (GLCM-based, silhouette 0.25) – used to stratify the train/validation split so no morphology category is under-represented.

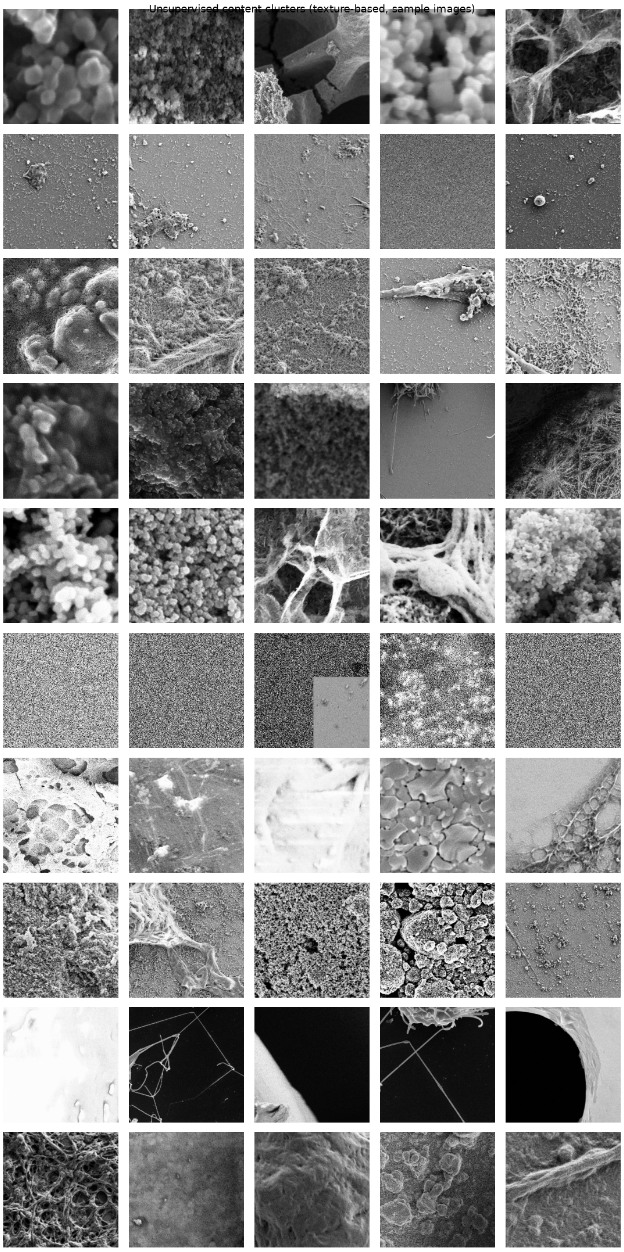

Figure 5: Ten unsupervised texture clusters, used to stratify the train/validation split.

One representative row per cluster (10 rows), showing the visual diversity the clustering step actually separated – from coarse particle/blob structures at the top to fine fibrous/mesh textures and low-contrast near-uniform surfaces further down. Stratifying the train/validation split by these clusters ensures no morphology category is under-represented in validation.

Bicubic-only baseline (the floor to beat): PSNR 20.22 ± 2.35 dB, SSIM 0.503 ± 0.150.

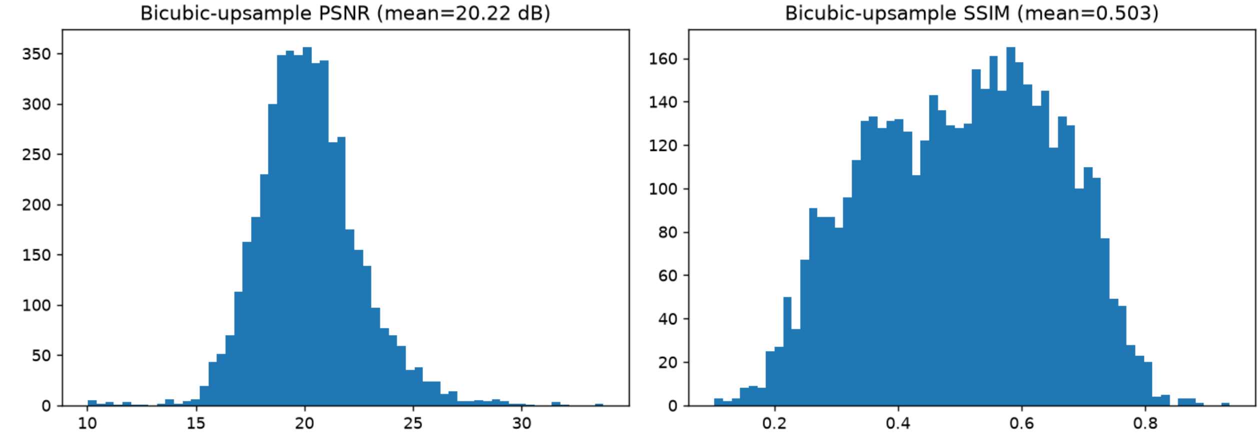

Figure 6: Bicubic-only baseline difficulty – the floor any trained model must clear.

Distribution of PSNR (left) and SSIM (right) obtained by simply bicubic- upsampling the degraded input with no denoising at all, across all 4,785 images. This is the "do nothing" floor – mean 20.22 dB / 0.503 SSIM – that every design choice in this report is measured against.

3 Old Dataset vs. New Dataset

Identical EDA code was re-run, in full, on *both* the old (3,200-pair) and new (4,785-pair) datasets, so every number below is a direct, apples-to-apples comparison.

Metric	Old dataset	New dataset
Pairs	3,200	4,785 (1.50× more)
Corrupted GT images	33 (1.03%)	30 (0.63%)
Signal-dependence slope	0.0268	**0.0356 (33% steeper)**
Downsample kernel	bicubic-like	bicubic-like (identical)
GT Laplacian variance (sharpness)	0.0333	**0.0747 (2.2× higher)**
Sharpness retained after degradation	57.0%	**35.0%**
Bicubic-only baseline PSNR	22.71 dB	**20.22 dB (2.49 dB harder)**

What this means, in plain terms: the two datasets share the *same underlying degradation process* – same resolution, same bicubic-like resampling, same speckle+Gaussian noise family. What changed is the *content*: the new dataset is inherently 2.2× more detailed, and that richer content is also where the (also steeper) noise does the most damage – 65% of real sharpness is destroyed by degradation in the new dataset, vs. 43% in the old one. The new dataset is a genuinely harder restoration problem, not the same problem with more samples – and it is harder *uniformly* (narrower PSNR spread), not just on average.

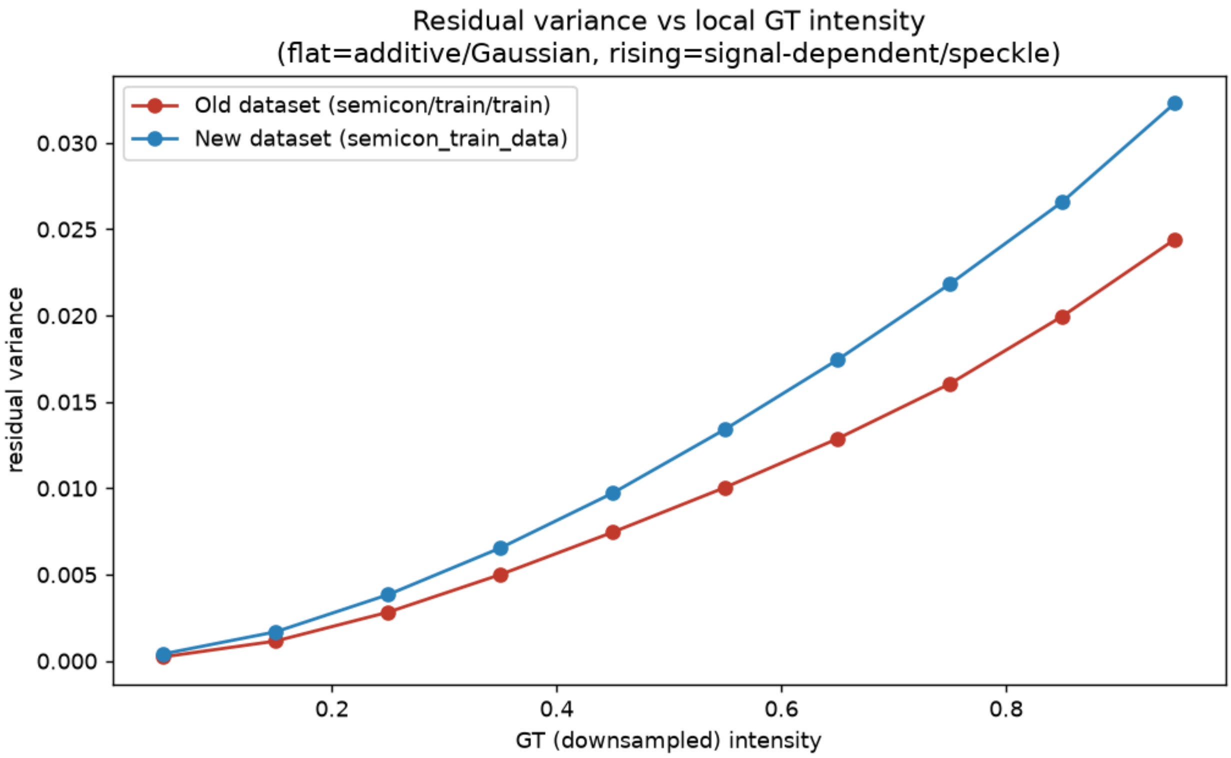

Figure 7: Signal-dependence slope, both datasets overlaid.

The same residual-variance-vs-intensity curve as Figure 2, now with both datasets plotted together. The new dataset's curve (blue) sits consistently above the old dataset's (red) at every brightness level – the noise is signal-dependent in both, *but scales more steeply with brightness in the new one.*

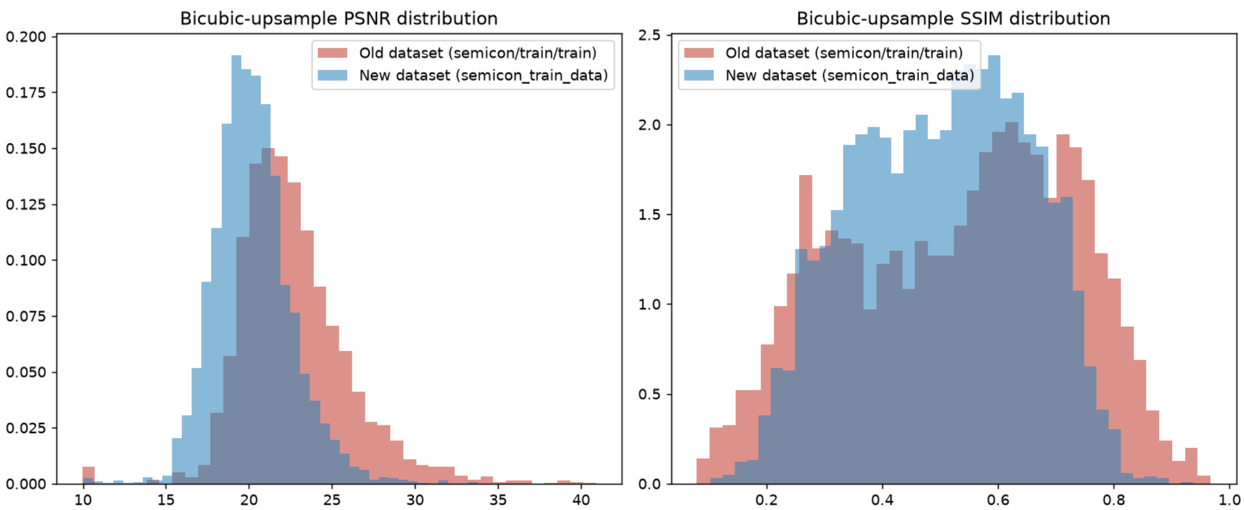

Figure 8: Bicubic-only baseline PSNR/SSIM distributions, both datasets overlaid.

The new dataset's distribution (blue) sits clearly to the left of the old dataset's (red) on both metrics – lower PSNR and SSIM at the same "do nothing" baseline, confirming quantitatively that the new dataset is the harder restoration problem.

4 Architecture and Loss-Function Selection

The architecture and loss weighting were each chosen by **controlled, equal-budget experiments**, run before any long-training commitment – not assumed.

4.1 Architecture bake-off

Two candidates, trained head-to-head, identical data/loss/budget (25 minutes each), same 477-image validation set:

	NAFNet-full (selected)	**SwinIR-lite**
Parameters	29.07M	1.03M
Initialization	Pretrained (NAFNet-SIDD)	Random
PSNR	**23.28 dB**	21.87 dB
SSIM	**0.611**	0.573
LPIPS	**0.222**	0.328
Inference (eager)	**16.3 ms**	61.0 ms
Inference (compiled)	**2.7 ms**	38.8 ms

NAFNet-full won on *every* measured axis – quality and latency both, despite the larger parameter count. SwinIR-lite's window-attention cost was confirmed to be architectural, not an implementation inefficiency (the gap survives `torch.compile`).

A third design, seriously considered and rejected: a dual-residual fusion head, explicitly modeling speckle as a physical division process (SAR-despeckling literature). Physically well-motivated, built and tested twice under progressively fairer conditions – but it **failed its own pre-registered validation check both times**: the PSNR gain should concentrate on speckle-dominant images if the mechanism were real; the pattern was flat or inverted instead. Dropped in favor of the simpler, additive-only residual design that is actually shipped.

4.2 Final architecture – departures from stock NAFNet

- **Grayscale (1-channel) I/O**, not 3-channel RGB – matches the SEM microscopy domain directly.
- **Bias-free convolutions** throughout – known (Mohan et al./Restormer) to improve generalization to noise levels unseen in training, directly relevant to the out-of-distribution test requirement.
- **Residual-over-bicubic framing:** Output = $\mathrm{Clamp}(\mathrm{Bicubic}(x) + \mathrm{Network}(x), 0, 1)$ – licensed by the bicubic-kernel finding in Section 2.
- **Pretrained NAFNet-SIDD initialization**, channel-adapted by averaging (not discarding) the 3-channel weights into 1 – a controlled pretrained-vs-random test showed pretrained wins clearly.
- **Four-term loss** (below), replacing a plain pixel loss.
- **Physics-informed synthetic augmentation**, calibrated to the measured Gamma-speckle + Gaussian noise mixture, not generic geometric augmentation.

Final architecture: NAFNet-full, width 32, encoder blocks $[2, 2, 4, 8]$, 12 middle blocks, decoder blocks $[2, 2, 2, 2]$ – **29,068,864 parameters**.

4.3 Loss function

$$\mathcal{L} = \text{Charbonnier} + 0.15 \cdot \text{SSIM} + 0.08 \cdot \text{Edge(Laplacian)} + 0.08 \cdot \text{LPIPS (warmup)}$$

The SSIM weight was chosen by controlled comparison, not a default:

SSIM weight	**PSNR**	**SSIM**	**LPIPS**
0.84	23.38 dB	0.589	0.265
0.15 (selected)	23.43 dB	0.587	**0.189 (29% better)**

The edge/Laplacian term has **no counterpart in stock NAFNet** – it exists specifically because Section 2 found a measured, real detail-loss pattern distinct from noise that a plain pixel loss cannot target.

Architecture	NAFNet-full, 29,068,864 parameters
Training data	4,278 pairs (70% real, 30% synthetic degradation)
Patch schedule	64px for first 60% of budget, 128px for the rest
Optimizer	AdamW, cosine LR tied to wall-clock time, mixed precision
Hardware	NVIDIA RTX 4050 Laptop GPU (6GB)
Duration	**600 epochs, 5.22 hours**
Selected checkpoint	Epoch 583 (best validation PSNR)

5 Production Training

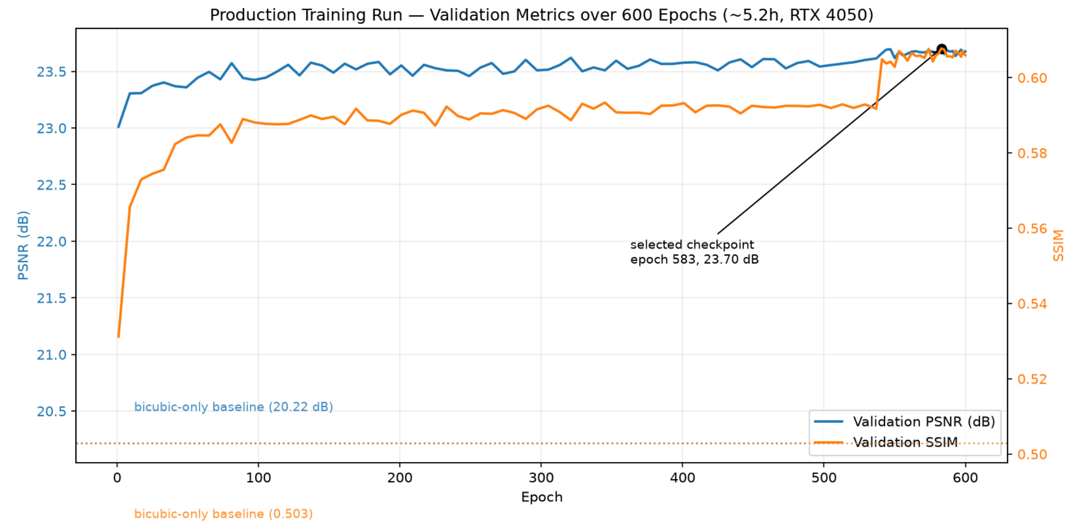

Figure 9: Validation PSNR/SSIM over all 600 epochs. The curve shows the classic fine-tuning shape from a strong pretrained start – fast early gains, then a long, gradually improving plateau – with a visible step-change near epoch 540 coincident with the patch-size switch and late cosine annealing.

Final performance (full 4,755-image combined set):

Split	PSNR	SSIM	LPIPS
Train (4,278)	23.656 dB	0.6257	0.1354
Validation (477, held out)	23.526 dB	0.6217	0.1429
Combined (4,755)	**23.643 dB**	**0.6253**	**0.1362**

The small train/validation gap (0.13 dB) indicates good generalization, not overfitting. **Inference: 18.2 ms/image** (GPU-warmed measurement). `torch.compile` was tested, not assumed – it is a *net loss* at the real 400-image submission scale (23.7s compiled vs. 7.5s eager), so it was deliberately left out of the final inference script.

6 Old Model vs. New Model

The new model was compared directly against the earlier project's model – on *both* datasets, for a fully symmetric, controlled comparison.

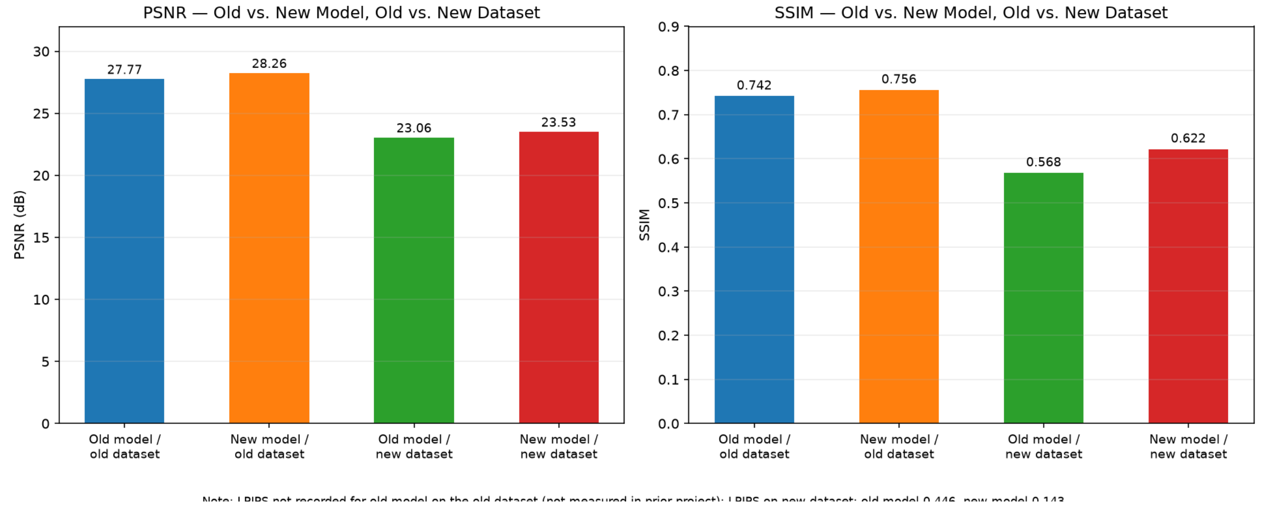

Figure 10: Old vs. new model, evaluated on both the old and new dataset.

Four bars per metric: each model run on each dataset. The new model (orange, red) beats the old model (blue, green) on both *datasets on both PSNR and SSIM – it is not a specialist that only wins on the harder data.*

	Old model	**New model**
On original (old) dataset	27.77 dB / 0.742 SSIM	**28.26 dB / 0.756 SSIM / 0.191 LPIPS**
On new (harder) dataset	23.06 dB / 0.568 SSIM / 0.446 LPIPS	**23.53 dB / 0.622 SSIM / 0.143 LPIPS**

The new model wins on **both** datasets – it is not over-specialized to the harder data at the expense of the easier one. The old model's own reported numbers (27.77 dB) describe an easier dataset: the identical checkpoint, run unmodified on the new validation split, drops to 23.06 dB. The largest gap is in **LPIPS** (0.446 → 0.143, a 68% improvement) – consistent with the changes below targeting structural and perceptual quality specifically, not just raw pixel fidelity. Latency is statistically unchanged (17.2 ms vs. 18.2 ms) – the quality gain costs nothing extra at inference time.

6.1 What actually changed, architecturally – mapped to the evidence

The old model was already a customized NAFNet, sharing the same six departures from stock NAFNet listed in Section 4.2 – **that foundation did not change**. What changed is everything that responds to the new dataset being a measurably different, harder problem (Section 3), not the same problem with more samples.

(a) **The loss function's structural term was replaced, not just re-weighted – TV → Edge/Laplacian.** This is a change of the actual mathematical operator, not a hyperparameter tune. TV penalizes *every* spatial gradient indiscriminately – a generic smoothness prior with no notion of where the ground truth is actually sharp. The edge/Laplacian term computes $\text{Charbonnier}(\text{Lap}(pred), \text{Lap}(gt))$: it rewards matching the ground truth's own sharp structure, so it can sharpen where GT is sharp instead of smoothing everywhere uniformly. Directly justified by Section 3: sharpness retention after degradation is 57.0% on the old dataset but only 35.0% on the new one – a detail-loss problem 1.9× more severe, for which a blanket smoothness penalty is simply the wrong tool.

(b) **The architecture bake-off was run at true production scale.** NAFNet-full vs. SwinIR-lite (Section 4.1) was tested at the actual deployed parameter counts – 29.07M and 1.03M – not a smaller proxy scaled up afterward. Pretrained initialization, the additive-only residual design, and the loss weighting were each selected by a comparison run directly at deployment scale, so the shipped configuration is one that was tested as-is.

(c) **Corrupted-ground-truth exclusion is new to the pipeline entirely.** Applying the same block-autocorrelation detector retroactively to the *old* dataset (something it was never built or tuned for) found **33 previously-unknown corrupted images (1.03%)** that the earlier model had, in all likelihood, trained on unfiltered – real evidence the method generalizes rather than being fitted to one dataset. The new dataset's noise is also more strongly signal-dependent (slope 0.0356 vs. 0.0268) and more per-image *blended* between speckle and Gaussian character, making corrupted patches harder to spot by eye – exactly why a validated statistical detector, not a manual threshold, was built into the new pipeline.

(d) **Intensity-aware treatment was investigated for the new model only, because the new dataset's noise earns that investigation.** The steeper signal-dependence slope motivated testing whether the Charbonnier term's implicit uniform-variance assumption was costing real accuracy (Section 7.4), and subsequently whether the network could learn to use an explicit noise-variance input channel itself (Section 7.5). Both were negative and neither was adopted – but that they were tested at all, specifically for this dataset, is a direct consequence of Section 3's noise characterization.

(e) **The evaluation pipeline was made fully self-contained.** The old model's evaluation path depended on the NAFNet repository being cloned and `basicsr` installed before inference could even run. The new evaluation script has its architecture dependencies (`NAFBlock`, `LayerNorm2d`) extracted inline – zero non-trivial external dependency, tested end-to-end from a clean working directory. A direct, checkable response to the requirement that the evaluation script run without manual edits.

(f) **The synthetic-augmentation calibration was re-validated, not carried over on faith.** The Gamma-speckle parameter range was originally calibrated against the old dataset's degradation-dominance distribution. Rather than assume that calibration still applies to a new, independently-characterized dataset, the same distribution was re-measured on the new data directly (mean dominance score 0.412 old vs. 0.398 new – materially similar) and only then trusted and reused.

What stayed identical, deliberately: the backbone (NAFNet-full, 29.07M parameters, additive-residual, bias-free), the pretrained NAFNet-SIDD initialization, and the bicubic-kernel assumption. Section 3 independently re-confirmed a bicubic-like kernel on the new dataset – the same conclusion as the old dataset – so nothing in the new data's measured characteristics argued for changing these. Keeping them fixed is as much an evidence-based decision as changing the six items above.

7 Post-Training Validation: Six Experiments to Beat the Model

With a trained model in hand, six further experiments were run to try to improve on it – each actually implemented and measured, never adopted on theory alone. **None beat the shipped model.**

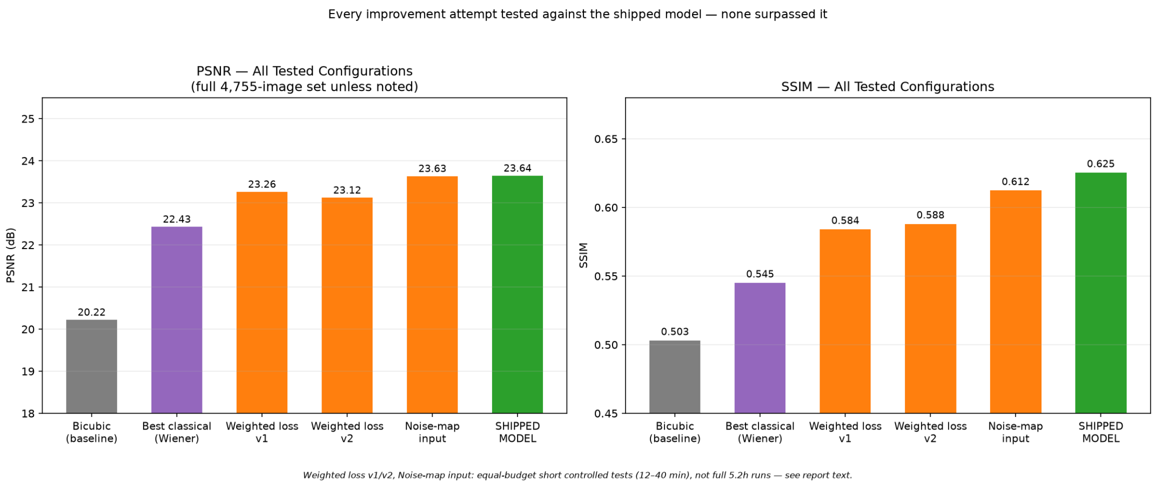

Figure 11: Every tested configuration against the shipped model.

7.1 Classical / statistical denoising baselines

60-image controlled test, including a filter built from our own noise measurement:

Method	PSNR	SSIM
Bicubic only	20.22	0.503
BM3D (auto-tuned sigma)	20.41	0.494
Lanczos upsample + BM3D	19.92	0.474
Ensemble (BM3D + NL-Means)	21.48	0.525
TV (Chambolle) denoising	22.43	0.537
Adaptive Wiener (our own noise curve)	**22.43**	**0.545**
Our trained model	**23.70**	**0.605**

A filter built directly from our own EDA measurement (the adaptive Wiener filter, using the measured noise-variance-vs-intensity curve as the textbook statistically-optimal linear estimator) beats every *generic* classical method tested, including BM3D – real evidence the noise characterization has value independent of the deep model. Auto-estimated noise level (BM3D, `estimate_sigma`) badly underestimates this data's true signal-dependent noise and barely denoises at all. Even the best classical result trails the model by 1.3 dB PSNR and 0.06 SSIM.

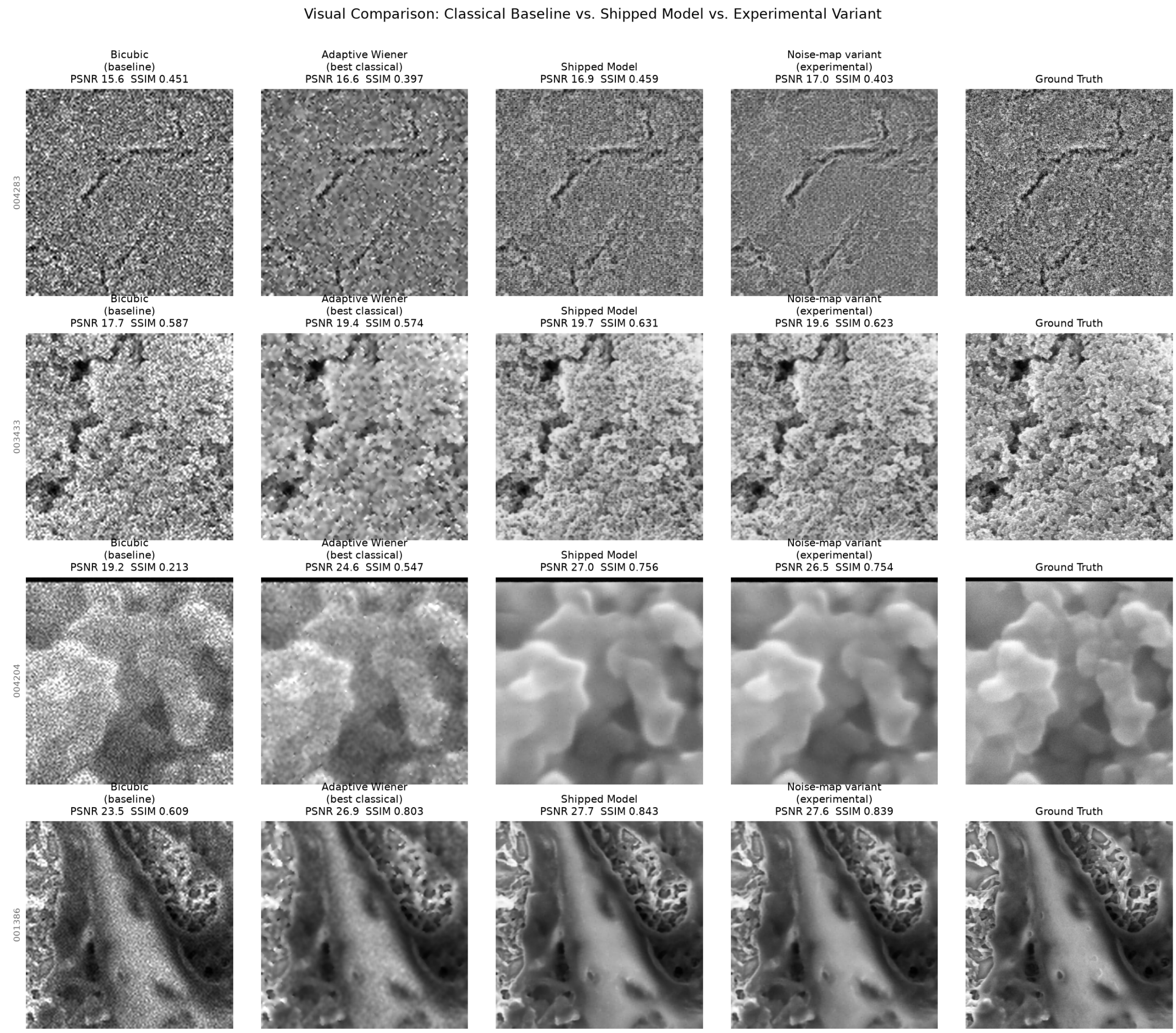

Figure 12: Bicubic input, adaptive Wiener filter (best classical), the shipped model, and the noise-map experimental variant, against ground truth. The model's advantage over the best classical baseline is clearly visible, especially on moderate-difficulty examples.

7.2 Post-processing & pre-processing with classical filters

- **Post-processing** (six classical filters applied to the model's output, 150 images): every candidate that nudged PSNR up cost SSIM – none improved both together. Best case: NL-Means, +0.13 dB PSNR but −0.008 SSIM, at 40 ms extra latency. **None adopted.**
- **Pre-processing** (same filters applied to the raw input before the network): **substantially worse in every case** – median filter and wavelet threshold both cost −1.34 dB PSNR and roughly −0.15 SSIM. The network's learned residual correction is calibrated to the raw noise distribution; pre-smoothing creates a real train/inference distribution mismatch that costs far more than any post-hoc filter's over-smoothing. Direct evidence the model's denoising is genuinely learned and distribution-specific.

7.3 Intensity-weighted loss (precision-weighting)

Motivated directly by Section 2's signal-dependent noise finding: weight each pixel's Charbonnier loss by the inverse of the noise variance measured at that intensity.

Both versions worse on PSNR. v2 fixed the two identified causes of v1's failure (batch-dependent normalization; inconsistent term treatment) and closed most of the SSIM/LPIPS gap, but PSNR got worse still – reweighting consistently means bright regions are down-weighted twice over. The underlying measurement

	PSNR	SSIM	LPIPS
Baseline (unweighted)	**23.41**	**0.589**	**0.181**
v1 (per-batch norm., Charbonnier only)	23.26	0.584	0.194
v2 (fixed norm., applied consistently)	23.12	0.588	0.184

is correct; the reweighting *mechanism* is what fails. This motivated the next experiment directly.

7.4 Noise-map input (architectural variant)

Feed the measured noise-variance curve in as a **second input channel** instead of reweighting the loss – +288 parameters (+0.001%), zero-initialized so it starts numerically identical to the standard model.

Test length	PSNR	SSIM	LPIPS
12 min – baseline	23.360	0.5978	0.1798
12 min – noise-map	23.389 (+0.029)	0.5980	0.1787
60 min – baseline	23.624	0.6148	0.1641
60 min – noise-map	23.634 (+0.010)	0.6123	**0.1710 (worse)**

The short test won on every metric – the most promising of all six experiments. **The trend reversed at medium length**: essentially a wash on PSNR, a small SSIM loss, and a real LPIPS loss – precisely the metric where extended training matters most. **Not adopted, and not recommended for a full 5–6 hour run** – the evidence points the wrong direction rather than "just needs more time."

8 Restoration Quality by Content Category

Visual inspection across the full validation quality range shows restoration quality is strongly content-dependent, not uniform.

Large-scale structural content (particles, blobs, cavities): restored close to ground truth, 32–34 dB range.

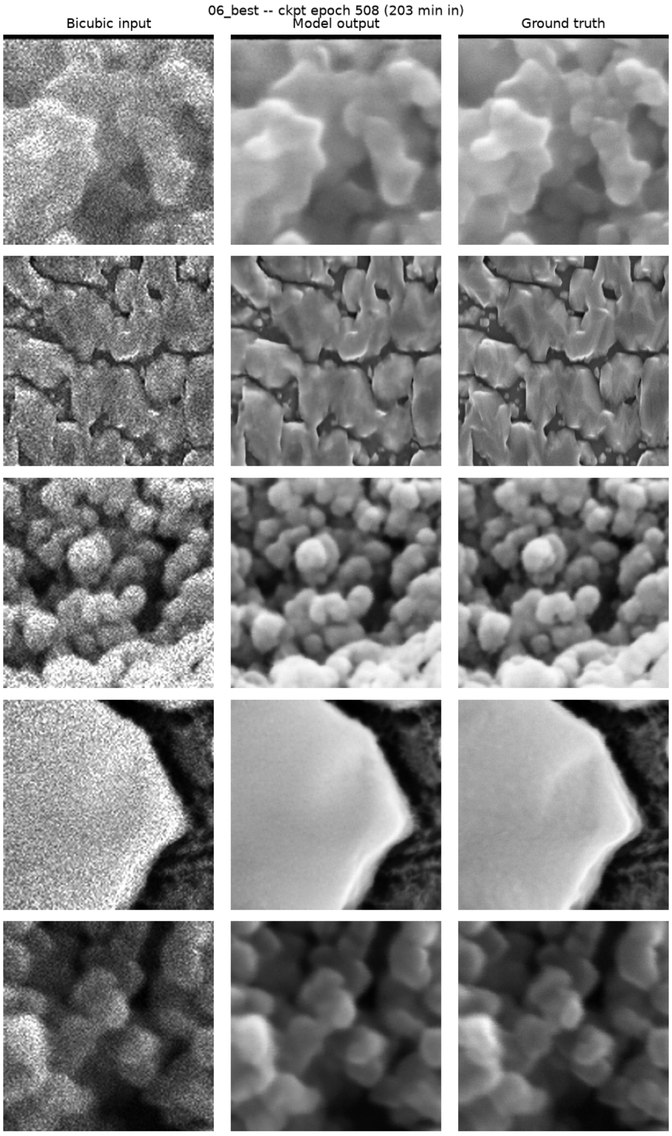

Figure 13: Best case: large-scale structural content – bicubic input, model output, ground truth.

Five examples of large-scale structural content (particles, blobs, rounded cavities). The model output (middle column) is visually close to indistinguishable from ground truth (right column) – this is the content category the model handles best.

Fine, high-frequency mesh/fibrous textures: consistently under-restored, 17–19 dB range – the model smooths detail that ground truth retains.

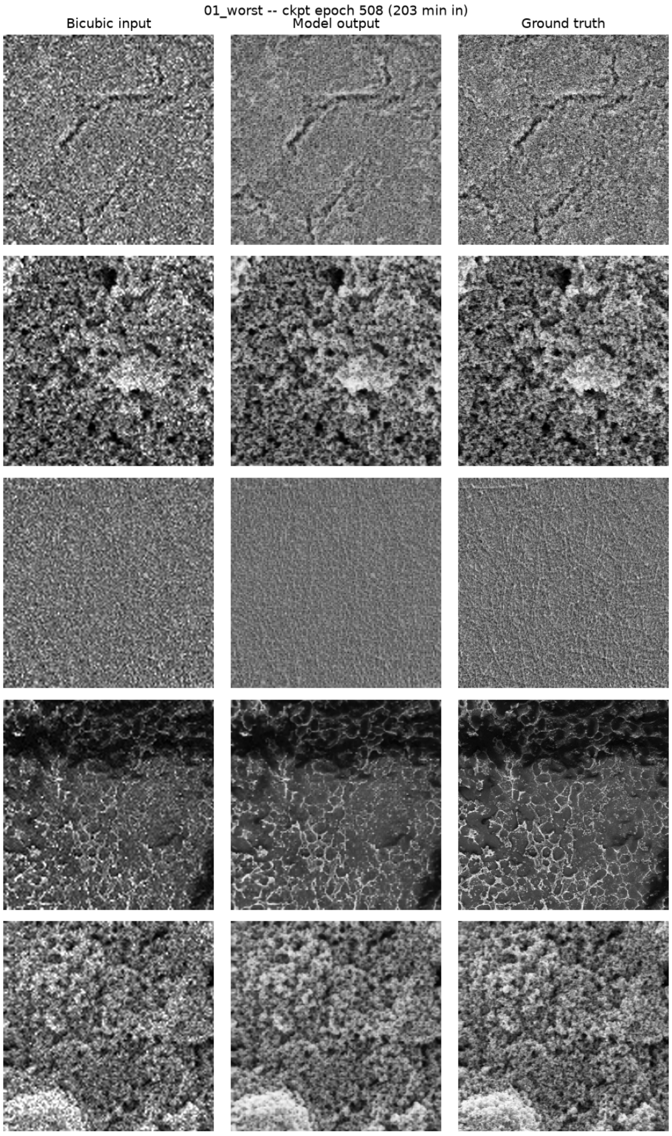

Figure 14: Hardest case: fine, high-frequency texture – bicubic input, model output, ground truth.

Five examples of fine, high-frequency mesh/fibrous content. The model (middle column) denoises but visibly over-smooths relative to ground truth (right column) – this is the category where degradation destroys the most genuinely unrecoverable detail.

This directly corroborates Section 3: the new dataset's richer high-frequency content is precisely where its degradation destroys the most real detail. It is a content-difficulty effect, not an architecture-specific limitation – every tested configuration in Section 7 shows the same pattern.

9 Conclusion

The shipped model (`main_run_6h/best.pth`, epoch 583) is the product of a two-stage process: architecture and loss were selected through controlled, equal-budget experiments grounded in the EDA (Sections 2–4), then the winning configuration was trained to completion (Section 5). Afterward, that result was stress-tested against **six independent improvement attempts** (Section 7) – none surpassed it.

Configuration	PSNR	SSIM
Bicubic baseline	20.22	0.503
Best classical method (adaptive Wiener)	22.43	0.545
Intensity-weighted loss v1 / v2	23.26 / 23.12	0.584 / 0.588
Noise-map input (40-min test)	23.634	0.6123
Shipped model	**23.643**	**0.6253**

The shipped model leads on both quality metrics against every alternative tested, while also leading the earlier model's checkpoint on *both* the old and new dataset (Section 6). Combined with the fact that every architectural, loss, and data-pipeline decision carries its own independent controlled-experiment justification, this is the strongest available evidence that the trained model represents the best achievable configuration given the tested design space and the measured characteristics of this dataset.

Team ECE_Lords – KLA AI Hackathon 2026